

A Two-Stage Modeling Approach to Risk Prediction with High-Dimensional Two-Phase Data

Jill S. Hasler^{1,2,†}, **Chenyu Bi**^{1,†}, Changcheng Li³, Ravi B. Parikh⁴, Weidong Ma¹, Jinbo Chen¹

¹ University of Pennsylvania, USA

² Fox Chase Cancer Center, USA

³ Dalian University of Technology, China

⁴ Emory University, USA

† Co-first authors



External Data: Improves Prediction but Often Incomplete

Patient	Age	Sex	SBP	...	Pain score	Fatigue score	...
 A	55	M	96		Missing		
 B	43	M	114		2	4	
 C	34	F	92		Missing		
...							

Complete EHR Data

Incomplete PRO Data

Advantages of combined data:

- EHRs provide rich data but miss granular patient information
- External data captures these missing details

⚠ However, external data is often only available for a subset of patients

Limitations of Common Approaches

Complete-case Analysis:

- Reduced effective sample size
- Distribution shift between the external-data subset and full cohort
- Limited generalizability to the target population

Imputation-based Analysis:

- Challenges with high-dimensional EHR predictors
- Instability under substantial missingness
- Limited accuracy under weak EHR–external data correlation

Need a method that uses all data without direct external-data imputation

Reframing the Problem as Two-Phase Data

Phase I:

Outcome	EHR		
Y	X_1	...	X_p
1	x_{11}	...	x_{p1}
0	x_{12}	...	x_{p2}
1	x_{13}	...	x_{p3}
...			

+

Phase II:

External Data		
Z_1	...	Z_q
NA	...	NA
z_{12}	...	z_{q2}
NA	...	NA

Similar to classical two-phase studies, but with new challenges:

- High-dimensional Phase I predictors (EHR data)
- Phase II availability is observational, not designed sampling

Existing two-phase methods do not readily address this setting

Our Two-Stage Modeling Framework

Outcome	EHR			External Data			Availability
Y	X ₁	...	X _p	Z ₁	...	Z _q	R
1	x ₁₁	...	x _{p1}	NA	...	NA	0
0	x ₁₂	...	x _{p2}	z ₁₂	...	z _{q2}	1
...							

Stage I: Use Phase I Data to Learn Initial Risk

$$\hat{\ell}(X) = \text{logit}\{\hat{P}(Y = 1|X)\}$$

- Use all patients
- Convert high-dimensional EHR predictors into one informative score
- Use sample-splitting to reduce overfitting

Stage II: Refine Prediction Using Phase II Data

$$\text{logit}\{P(Y = 1|X, Z)\} = \beta_0 + \beta_1 \hat{\ell}(X) + \beta_2^T Z$$

- Use patients with both Phase I and II data
- Similar to model recalibration and revision

Model Fitting – Selection Bias Correction

Stage I: $\hat{\ell}(X) = \text{logit}\{\hat{P}(Y = 1|X)\}$

Stage II: $\text{logit}\{P(Y = 1|X, Z)\} = \beta_0 + \beta_1 \hat{\ell}(X) + \beta_2^T Z$

The estimated parameters β can be obtained (Breslow and Cain, 1988) :

$$\sum_{i=1}^N R_i(1, X_i^T, Z_i^T)^T \left\{ Y_i - \frac{\exp\{\text{logit } P(Y = 1|X_i, Z_i) + \hat{o}(X_i)\}}{1 + \exp\{\text{logit } P(Y = 1|X_i, Z_i) + \hat{o}(X_i)\}} \right\} = 0$$

where

$$o(x) = \log \left\{ \frac{P(R = 1|Y = 1, X)}{P(R = 1|Y = 0, X)} \right\}$$

Model Fitting – Selection Bias Correction

To model R, the availability of phase II data:

$$\text{logit}\{P(R = 1|Y = y, X)\} = \alpha_y^T X_r + \delta_y^T B\{\hat{P}(Y = 1 - y|X)\}$$

- X_r : selected variables predictive of R
- $\hat{P}(Y = 1 - y|X)$: residual term captures information missed by stage I
- $B(\cdot)$: B-spline basis, which allows flexible modeling of the selection mechanism

Why this helps?

- Subjects poorly predicted by stage I tend to be more informative in two-phase sampling. (Fithian and Hastie, 2014; Shen et al., 2021)

Approximation for Estimating AUC

Use inverse-probability weighting to estimate the risk score $S(X, Z) = P(Y = 1|X, Z)$ distribution under selective phase II observation.

Under our framework, it is reasonable that $S(X, Z)$ is well calibrated:

$$P\{Y = 1|S(X, Z)\} \approx S(X, Z)$$

Weighted empirical estimator:

$$\hat{F}(x, z; \hat{\beta}, \hat{\tau}, \hat{l}) = \sum_{i=1}^N N^{-1} \frac{I\{R_i = 1, \hat{l}(X_i) \leq \hat{l}(x), Z_i \leq z\}}{\sum_{y=0}^1 P(R_i = 1|Y_i = y, X_i; \hat{\tau}) p_y\{\hat{l}(X_i), Z_i; \hat{\beta}\}}$$

Approximate AUC:

$$\widehat{AUC}(X, Z; \hat{\beta}, \hat{\tau}, \hat{l}) = \int TPR_v(X, Z; \hat{\beta}, \hat{\tau}, \hat{l}) dFPR_v(X, Z; \hat{\beta}, \hat{\tau}, \hat{l})$$

Simulation Setup

Data Generation

- $N = 10,000$
- # EHR (X): 300 with 10 truly predictive
- # external (Z): 3
- $(X, Z) \sim$ multivariate standard normal distribution with correlation $\rho^{|i-j|}$, $\rho = 0, 0.3$ or 0.5
- Y with $P(Y=1) = 0.2$:

$$\text{logit}\{P(Y = 1|X, Z)\} = \phi_0 + \phi_1 X + \phi_2 Z$$

- R with $P(R=1) = 0.05$ or 0.2 :

$$\text{logit}\{P(R = 1|X)\} = \tau_0 + \tau_1 \frac{\sum_{j=1}^q X_j}{\sqrt{q}}$$

- Stage I Model: adaptive lasso with two-fold sample-splitting procedure
- Simulation times: 1,000
- Methods compared: Proposed method, Benchmark (full Z observed), Complete-case (CC) analysis

Simulation Results

Table 1. Results on the estimation of coefficients and AUC, where the stage I model is fitted using adaptive lasso. Est” refers to the mean coefficient/AUC estimates, “SE” to the empirical standard error, and “Asy SE” to the averaged asymptotic standard error. “CP” refers to the 95% coverage probabilities calculated compared to two-stage benchmark estimates.

ρ	Coefficient / AUC	P(R = 1) = 0.05		P(R = 1) = 0.2	
		Est (SE / Asy SE)	CP	Est (SE / Asy SE)	CP
0	Intercept	-0.066 (0.086/0.080)	0.966	-0.065 (0.051/0.039)	0.981
	$\hat{I}(X)$	1.120 (0.075/0.069)	0.971	1.108 (0.047/0.032)	0.974
	Z_1	0.408 (0.136/0.139)	0.952	0.400 (0.070/0.069)	0.968
	Z_2	0.695 (0.148/0.146)	0.932	0.683 (0.071/0.072)	0.968
	Z_3	-0.585 (0.143/0.144)	0.960	-0.579 (0.070/0.070)	0.977
	AUC	0.854 (0.014/0.014)	0.943	0.852 (0.008/0.008)	0.982

- Coefficient estimates from the proposed method were close to the benchmark
- Asymptotic standard errors closely matched empirical standard errors
- Coverage probabilities were close to nominal levels
- Similar conclusions were obtained in other settings

Simulation Results

Proposed vs CC

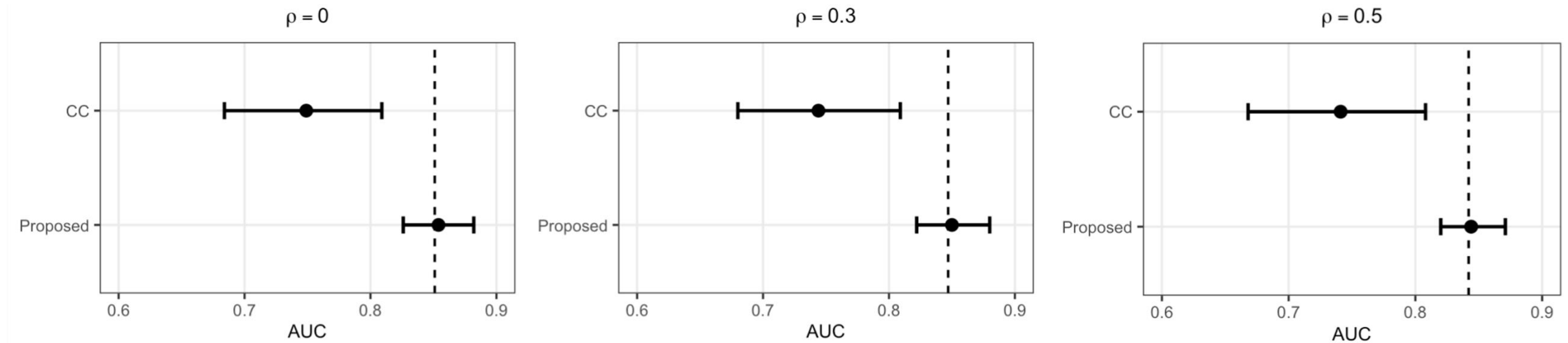


Figure 1. Comparison of AUC estimation between the proposed method and the CC approach when $P(R = 1) = 0.05$. “AUC” refers to the mean AUC estimates, and the dashed line represents the corresponding benchmark value.

- Proposed estimates closely matched benchmark values
- Much smaller variability than CC analysis

Real Data : Oncology Risk Prediction

- **Cohort:** 8,555 oncology patients from University of Pennsylvania Health System (UPHS)
- **Outcome:** 180-day mortality (485 deaths; 5.7%)
- **EHR predictors:** 204 variables
- **PRO predictors:** 12 variables (55% available)

Preliminary Model:

Lasso: $R \sim X$

- Several variables were selected, but effects were small
- No single EHR variable strongly drove PRO data availability

Stage I Model:

Adaptive lasso: $Y \sim X$

Stage II Model:

Logistic: $Y \sim \hat{l}(X) + Z$

- **Two-Stage I:** Model R using $B(\cdot)$ only
- **Two-Stage II:** Model R using $B(\cdot)$ and X_r
- X_r : 10 variables with the largest coefficients in the preliminary model

Real Data : Oncology Risk Prediction

Two-Stage I:

- (1) Adaptive lasso: $Y \sim X$
 - (2) Logistic: $Y \sim \hat{l}(X) + Z$
 - Model R using $B(\cdot)$ only
- Use our proposed method

Two-Stage II:

- (1) Adaptive lasso: $Y \sim X$
 - (2) Logistic: $Y \sim \hat{l}(X) + Z$
 - Model R using $B(\cdot)$ and X_r
- Use our proposed method

Two-stage CC:

- (1) Adaptive lasso: $Y \sim X$
 - (2) Logistic: $Y \sim \hat{l}(X) + Z$
- Only use patients with both EHR and PRO data

Table 2. Coefficient and AUC estimates for three models predicting short-term mortality

	Two-Stage I	Two-Stage II	Two-Stage CC
Intercept	-0.695 (-1.624, 0.235)	-0.675 (-1.601, 0.252)	-0.115 (-1.050, 0.820)
$\hat{l}(X)$	0.713 (0.643, 0.782)	0.707 (0.636, 0.779)	1.000 (0.857, 1.143)
Performance Status	0.379 (0.208, 0.550)	0.383 (0.213, 0.553)	0.306 (0.126, 0.487)
Nausea Frequency	0.119 (-0.021, 0.259)	0.117 (-0.024, 0.258)	0.128 (-0.020, 0.276)
Numbness	-0.194 (-0.342, -0.046)	-0.197 (-0.345, -0.050)	-0.185 (-0.341, -0.029)
Quality of Life	0.216 (0.045, 0.388)	0.221 (0.051, 0.392)	0.154 (-0.020, 0.329)
AUC	0.820 (0.796, 0.845)	0.821 (0.796, 0.846)	0.867 (0.818, 0.915)

Two-stage I/II differed from Two-Stage CC

- Distributional differences in X between $R = 1$ and $R = 0$ groups

Two-Stage I/II gave nearly identical estimates

- The $B(\cdot)$ flexibly accounted for phase II data availability

References

Breslow, N. E. and Cain, K. C. (1988). Logistic regression for two-stage case-control data. *Biometrika* 75, 11–20.

Fithian, W. and Hastie, T. (2014). Local case-control sampling: Efficient subsampling in imbalanced data sets. *The Annals of Statistics* 42, 1693–1724.

Shen, X., Chen, K., and Yu, W. (2021). Surprise sampling: Improving and extending the local case-control sampling. *Electronic Journal of Statistics* 15, 2454 – 2482.